CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

FAMILY TREE

Aim:

To implement a family tree using Prolog and derive relationships between family members.

Objective:

- To represent parent-child relationships.
- To derive father, mother, sibling and grandparent relationships.
- To demonstrate rule-based reasoning.

Algorithm:

- **Start.**
- **Define parent facts.**
- **Define male and female facts.**
- **Define rules for father, mother, sibling and grandparent.**
- **Query the required relationship.**
- **Display the result.**
- **Stop.**

Flowchart:

START

|

v

Define Family Facts

```
  |
  v
Define Relationship Rules
  |
  v
Enter Query
  |
  v
Match Facts and Rules
  |
  v
Display Relationship
  |
  v
STOP
```

Prolog Program:

```
% Family Tree

male(ramesh).
male(arun).
male(kiran).
```

```
female(sita).
female(priya).
female(neha).

parent(ramesh, arun).
parent(sita, arun).

parent(ramesh, priya).
parent(sita, priya).

parent(arun, kiran).
parent(priya, neha).

father(X, Y) :male(X),
   parent(X, Y).

mother(X,    Y)
   :female(X),
   parent(X, Y).

grandparent(X, Z)
   :parent(X, Y),
   parent(Y, Z).
```

```
sibling(X,  Y)  :-
    parent(P,  X),
    parent(P, Y),
    X \= Y..
```

Output:

```
82 ?- father(john, robert).
true.

83 ?- mother(mary, robert).
true.
```

Result:

Thus, the family tree is successfully implemented using Prolog facts and rules.